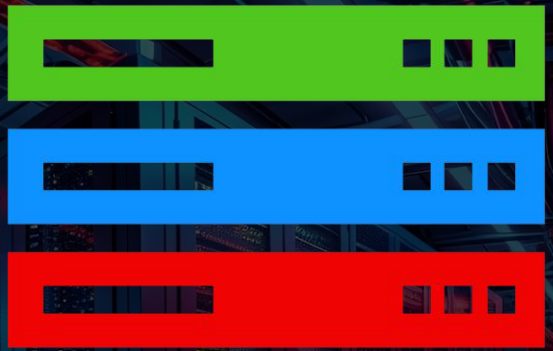


HPSF

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SOFTWARE FOUNDATION

Building the Future of HPC Software, **Together!**

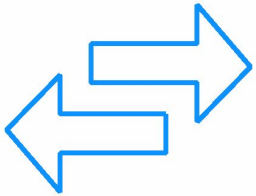
Julien Bigot, Xavier Delaruelle & Axel Huebl

HPC Best Practices webinar 2026-10-28

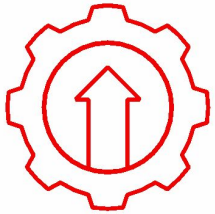
The **H**igh **P**erformance **S**oftware **F**oundation?



Performance



Portability



Productivity

A **neutral** hub for **open source**, **high performance** software

In 2023, ECP was ending,
efforts like NumPEx, or Fugaku-next were starting
… How to sustain the software stack, together?



launched at ISC 2024

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#Neutral

#Open-source

#Sustainable

#High-performance

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Neutrality as a way to sustainability

Solving the “chicken & egg” problem of **sustainability**

Sustainability in OSS is ensured by large & diverse contributor base

1. **Contributors** emerge from the **community** and require **trust**
   - No one organization can **control** the direction of the project
   - Projects are **open** to new contributors and new ideas
2. **Communities** emerge around **good products** and **trust**
   - Projects are and will remain **usable by anyone**
3. **Good products** require enough **contributors** to build them

Neutral, **open governance** ensures that we can build the **broadest** possible **communities**

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Trust in a sustained open-source strategy

- Projects write an **open-source project charter**: clarify governance model
  - Who leads the project? What are the process to select new leaders?
  - The project is and will remain committed to open-source
- **Transfer the name** and associated rights to LF
  - Brand, trademark, website, domain name, github / public forge co-ownership, ...
- **Code ownership remains with its authors**
  - Under an OSI-compliant open-source licence
- LF **gives back the usage right** on everything transferred
  - Acts as a trusted third party
  - Ensures the charter is and will remain followed, that the project is committed to open-source
  - Enforces that no `${myproject}`-enterprise tries to leach on the project popularity
  - Offers last resort ownership to ensure transition if all project owners evaporate

HPSF project lifecycle as a path to sustainability



Emerging

- Libre, open source, vendor-neutral HPC project
- Committed to open governance
- Working towards best practices

Open-source & committed to move forward

Established

- Wide usage by at least 3 organizations
- Steady development
- Robust development practices & processes, open to new developers

Robust for production use & committed to move forward

Core

- Steady development from multiple organizations
- No single organization has a majority in governance
- Sustainable cycle of development and maintenance

No single institution is the bottleneck

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Fund & vote

Members

Premier $175k/y

Lawrence Livermore National Laboratory, Microsoft, Sandia National Laboratories

General $5k-50k/y

AMD, Argonne National Laboratory, arm, aws, cea, cscs, HPE, Fujitsu, kitware Delivering Innovation, Los Alamos National Laboratory EST. 1943, NVIDIA, Oak Ridge National Laboratory, RIKEN, R-CCS

Associate for Academia

Berkeley Lab, सी डैक CDAC, FORTH, Jülich Forschungszentrum, Tennessee Tech, UCL, Universität der Bundeswehr München, University of Oregon, UTokyo

Governing Board

Lawrence Livermore National Laboratory, Microsoft, Sandia National Laboratories, cea, kitware Delivering Innovation, WARPX, kokkos

Participate & vote

Technical Advisory Council

Join & vote

WGs & committees

- Outreach
- Events
- CBench
- CI/CD
- Binaries
- Workforce
- ...
- ...

Projects

Spack, Charliecloud, AMReX, Apptainer, E4S, Environment Modules, Open CHAMI, CHAPEL, HPX, VISKORES, kokkos, HPCToolkit, flux, Trilinos, Ramble, wi4mpi, WARPX, ClusterShell

Who are we?

Julien Bigot:

- Representative of CEA (France Atomic Energy Commission)
- Elected at the governing board by general members



Xavier Delaruelle

- TAC representative for Environment Modules project



Axel Huebl

- TAC representative for WarpX project
- Elected at the governing board by the TAC



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Membership

Americas Europe Asia

Premier

Lawrence Livermore National Laboratory

Microsoft

Sandia National Laboratories

General

AMD

aws

Argonne NATIONAL LABORATORY

arm

cea

cscs

FUJITSU

HPE

kitware Delivering Innovation

Los Alamos NATIONAL LABORATORY EST.1943

NVIDIA

OAK RIDGE National Laboratory

RIKEN R-CCS

Associate

BERKELEY LAB

सी डैक CDAC

FORTH

JÜLICH Forschungszentrum

Technical University of Munich

UCL

Universität der Bundeswehr München

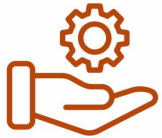
UNIVERSITY OF OREGON

UTokyo

Tennessee TECH

HPSF HIGH PERFORMANCE SOFTWARE FOUNDATION

What's in it for members?







HPC Providers (HW/SW/Services)

- **Leverage HPSF projects** to enhance your services and products
- **Ensure your products are well supported** by HPSF software
- **Secure mindshare** and collaborate with some of the leading software teams in the HPC space

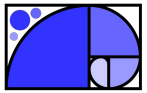
HPC Users (Scientists, Analysts)

- **Leverage HPSF projects** to develop, build, deploy and profile your projects
- **Connect with a community** that can help you use the latest high performance computing software and hardware
  - CPUs, GPUs, AI/ML architectures
- **Voice concerns and requirements** to the HPSF community
- **De-risk software decisions** knowing there's a community to rely on

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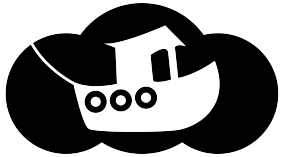
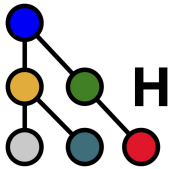
Projects


ClusterShell


AMReX

Spack
wi4mpi

RAMBLE

APPTAINER
OpenCHAMI

Charliecloud

HPCToolkit

ENVIRONMENT MODULES

E4S

kokkos

HPX

VISKORES
TRILINOS

WARPX

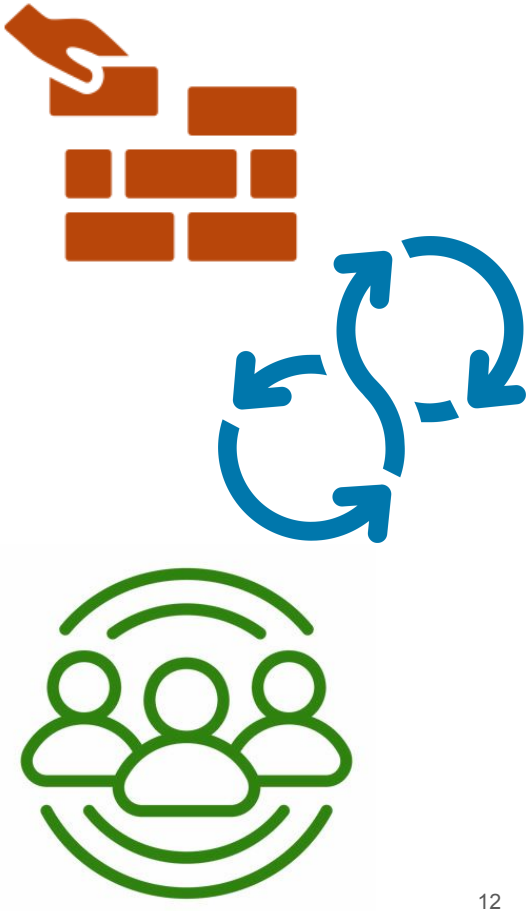
flux

CHAPEL


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What's in it for projects?

- **Build a community**
  - grounded in **neutral open source governance**
- **Share knowledge** and **find synergies** with others
  - Improve your governance model and learn from the most advanced projects
  - CI, software engineering best practices, community upkeep
  - Participate in working groups that aim to bring HPC open source software to the wider computing world
- **Get support** for all the "boring" aspects
  - Marketing, legal, collaborative tools, logos & website design, meetings organization, event promotion, etc.



The LF is a broad scope foundation

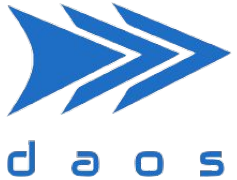
Being part of this ecosystem make collaboration beyond HPSF easier



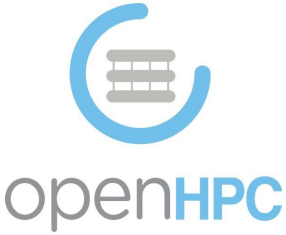












Expanding the HPC Workforce: HPSF & Sustainable Research Pathways

- **Strategic Partnership:** Leveraging Sustainable Horizons Institute (SHI) to bridge the gap between students and HPSF open-source projects.
- **Mentorship Focus:** Connecting scientists, mathematicians, and engineers with expert HPSF contributors.
- **Proven Pipeline:** Successful matching completed in December 2025; inaugural cohort begins **Summer 2026**.

SUSTAINABLE HORIZONS INSTITUTE

HPSF
HIGH PERFORMANCE SOFTWARE FOUNDATION

in collaboration with:

U.S. DEPARTMENT OF ENERGY

NSF

CASS

PESO

NAIRR Pilot National Artificial Intelligence Research Resource Pilot

HPSF Conference Last year in Chicago

HPSFcon plenary

- Building & Sustaining Communities
- Emerging Technologies & the Future of HPC
- Productivity, Performance, & the HPC Pipeline
- Project & Ecosystem Growth
- Trust, Security, & Open Collaboration
- User Experiences

Project Working Sessions

- **Programming systems:** HPX, Kokkos
- **Deployment:** Apptainer, Charliecloud, Spack, Environment Modules
- **Libraries**: AMReX, Trilinos
- **System management:** OpenChami, Flux Framework

Join us this year in Montreal!
April 12 - 16



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HPSF Community Summit in Europe



- **Last year**: February 25 - 27, 2026
- **Location**: TU Braunschweig, Germany

A platform in Europe for users, developers and members of HPSF projects to:

- learn about recent developments,
- report on their use cases and experiences,
- interact with core developers and the community,
- form a network of HPSF project users and developers in Europe.

Expect a new announcement for fall 2027

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Committees & Working Groups: Driving the Ecosystem

CI/CD

Providing access to GPUs and next-gen architectures

Leads/Contact:
Alec Scott & Zack Galbreath

Binaries

Reducing sysadmin burden via ABI compatibility and secure binary signing

Leads/Contact:
Ryan Krattiger

Benchmarking

Developing reproducible performance benchmarks for diverse software stacks

Leads/Contact:
Olga Pearce & Jayesh Badwaik

Outreach Committee

Drive community outreach to boost HPSF & Project visibility.

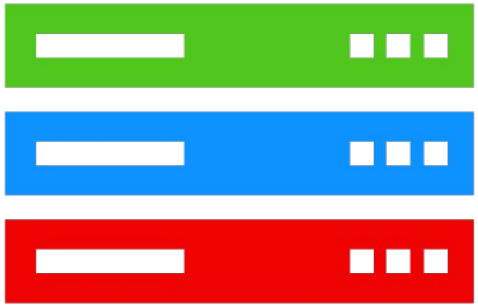
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Conclusion

- HPSF is a **gathering of projects**
- Supported by **member** institutions

- HPSF helps get some **trust** on the long term **viability** of these projects
- It also offers them **support** to develop and grow

- But the real value is in collaboration
  - **Let's hear from actual project representatives**



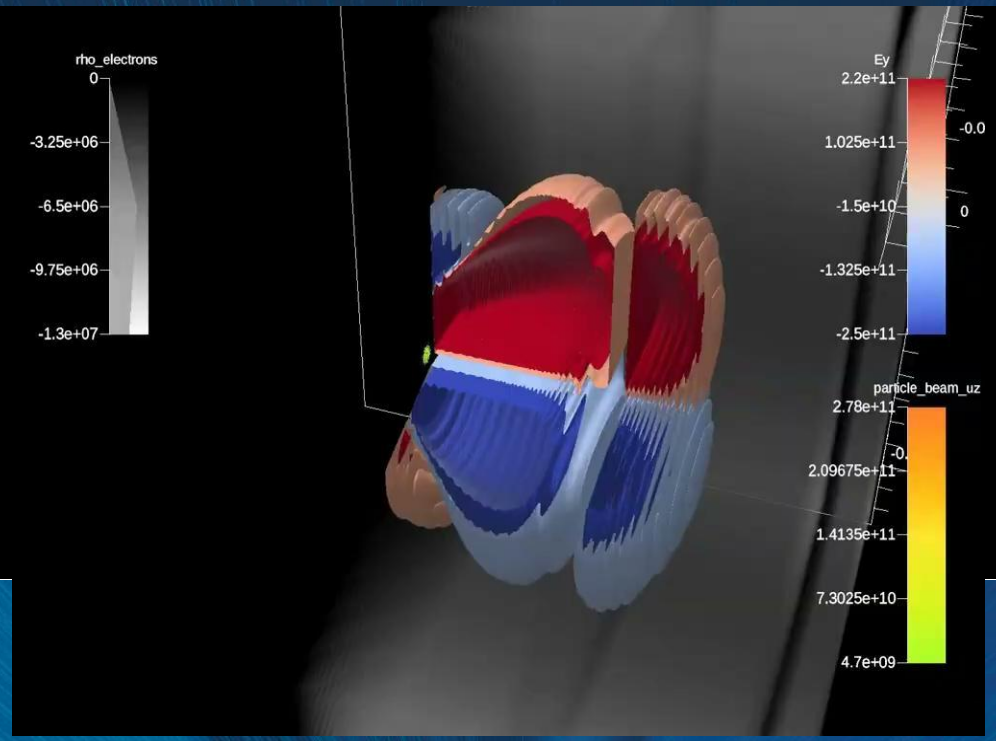
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hpsf.io

The HPSF: Building the Future of High-Performance Software, Together – WarpX Project

Axel Huebl for the BLAST WarpX team and collaborators at
LBNL, LLNL, CEA-LIDYL, SLAC, DESY, CERN, Helion Energy, Avalanche, TAE, Realta Fusion

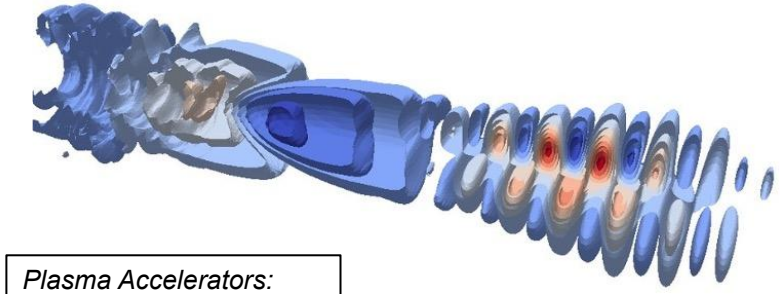


multi-stage LPA simulation in a boosted frame with WarpX
transversely focusing fields & beam

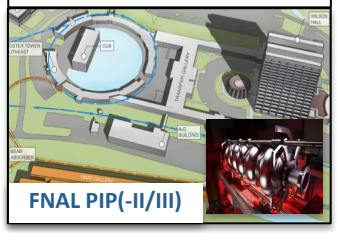
HPC Best Practices Webinars

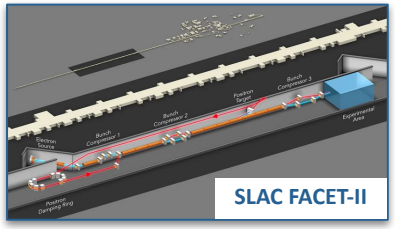
Wednesday, Oct 28, 2026
virtual / USA

Visualization of Selected Applications

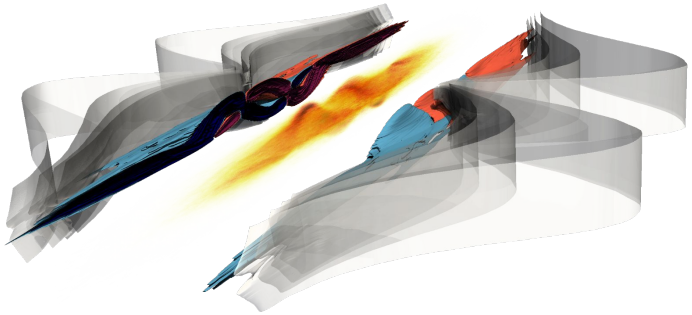


Radio-Frequency Accelerators

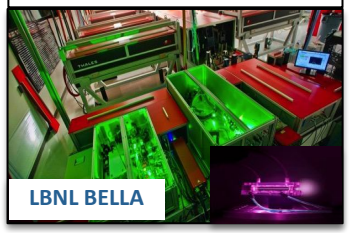




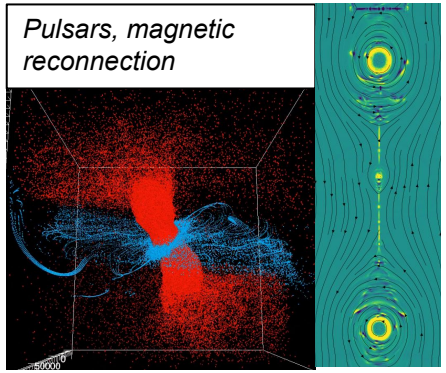
High-Energy Physics: Interaction Point (Beam-Beam)



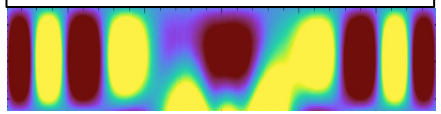
Plasma Accelerators: Laser- or Beam-Driven



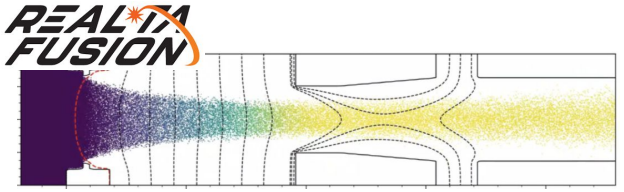
Pulsars, magnetic reconnection



Microelectronics

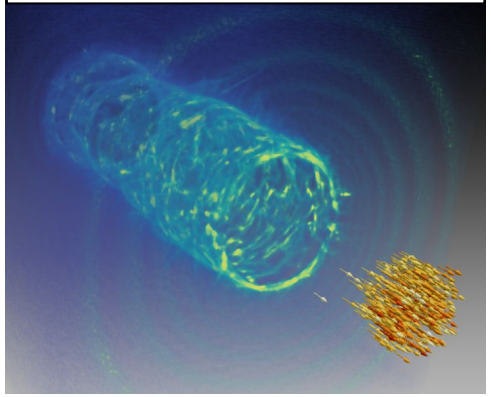


Alternative/Commercial/Startup Fusion Schemes

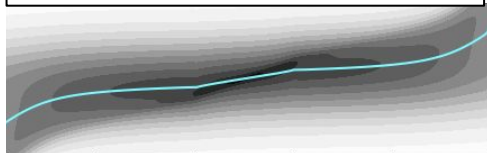


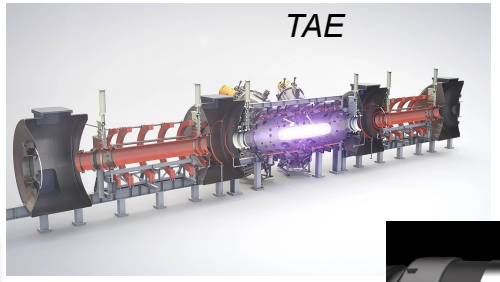


Laser-Plasma Interaction, Laser-Ion acceleration, High-Energy Density Physics

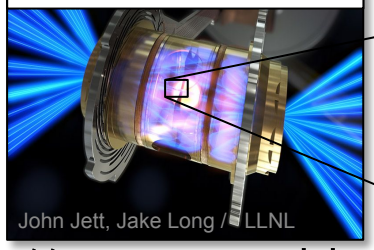


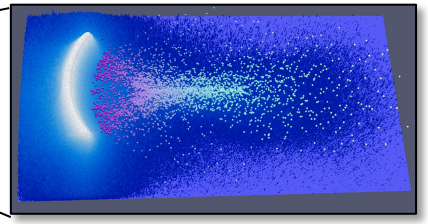
Magnetic Fusion Sheets

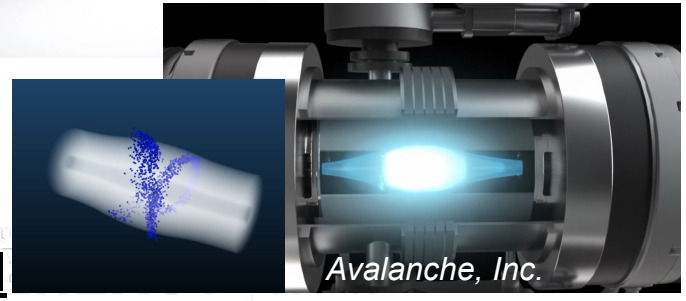




Kinetic effects in Inertial Confinement Fusion



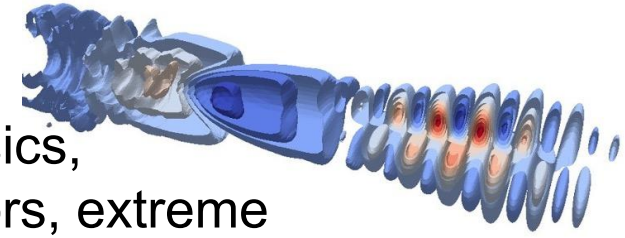




https://warpx.readthedocs.io/en/latest/highlights.html

WARPX a Community PIC Code for Exascale

Applications

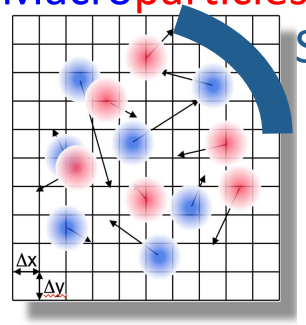


laser-plasma physics,
particle accelerators, extreme
light sources, fusion devices & plasmas, …

Particle-in-Cell

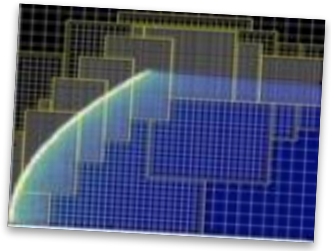
- electromagnetic or electro/magnetostatic
- 1-3D, RZ+, spherical
- time integration: explicit, implicit

Macroparticles



electromagnetic (EM) fields on a grid

adaptive mesh-refinement (AMR)



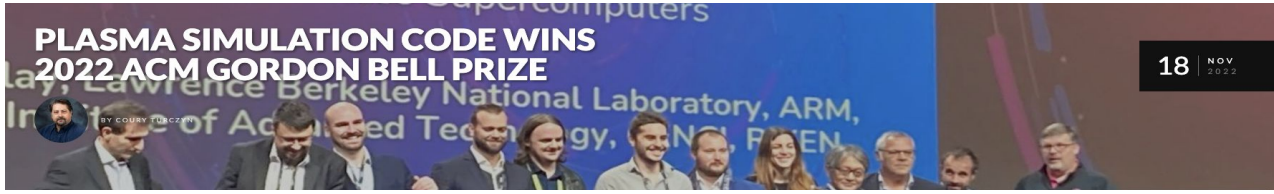
International Contributors incl. private sector

BERKELEY LAB · cea PARIS-SACLAY · DESY. · LLNL · loa · CERN · HELION · UR LLE · SLAC · AVALANCHE · tae TECHNOLOGIES · REALTA FUSION

open source initiative®

THE LINUX FOUNDATION
HIGH PERFORMANCE SOFTWARE FOUNDATION

Award–Winning Code & Science



Detailed Physical Models

- Full documentation, benchmarks, examples
- Easy-to-use boosted frame
- collisional, quantum & fusion processes
- PIC-fluid hybrid, *and much more*

Portable, Multi-Level Parallelization

- GPUs & CPUs
- Desktop to supercomputer

BLAST BEAM PLASMA & ACCELERATOR SIMULATION TOOLKIT

Scalable & Standardized

- Python APIs, schemas
- In situ processing
- Open community ecosystem

openPMD · LASY · AMReX · picmi Standard

J-L Vay et al., DOI:10.5281/zenodo.4571577, blast-warpx.github.io (2018)
L Fedeli, A Huebl et al., SC22, DOI:10.1109/SC41404.2022.00008 (2022)

BLAST
BEAM PLASMA & ACCELERATOR SIMULATION TOOLKIT

Developed by an international, multidisciplinary team

WarpX open governance

open source initiative®

THE LINUX FOUNDATION

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BERKELEY LAB

ATAP ACCELERATOR TECHNOLOGY & APPLIED PHYSICS DIVISION

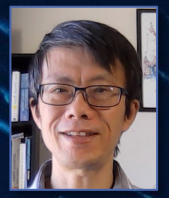
AMCRD

NERSC

Jean-Luc Vay



Ji Qiang



Arianna Formentii



Axel Huebl



Rémi Lehe



Chad Mitchell



Olga Shapoval



Edoardo Zoni



David Grote



Justin Angus



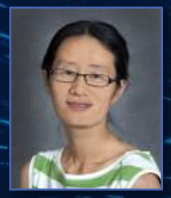
Ann Almgren



Kevin Gott



Junmin Gu



Revathi Jambunathan



Andrew Myers



Weiqun Zhang



UR LLE

Kale Weichmann



Roelof Groenewald



HELION

REALTA FUSION

Eric Clark



Prabhat Kumar



Germany

DESY.

Maxence Thévenet



Alexander Sinn



Ángel Ferran Pousa*



Rob Shalloo



France

loa

Igor Andriyash



Switzerland

CERN

Lorenzo Giacomel*



Severin Diederichs



SLAC

Lixin Ge



France

DE LA RECHERCHE À L'INDUSTRIE

cea PARIS-SACLAY

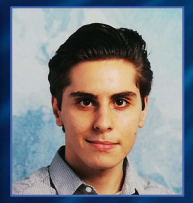
Henri Vincenti



Luca Fedeli



Thomas Clark



Pierre Bartoli*



CASUS CENTER FOR ADVANCED SYSTEMS UNDERSTANDING

Franz Poeschel



*former members

tae TECHNOLOGIES

Ryan Sandberg



AVALANCHE

Marco Garten



over 120 **contributors,** incl. from the private sector

Reuse: In Applications like WarpX, *"Everything" Comes Together*





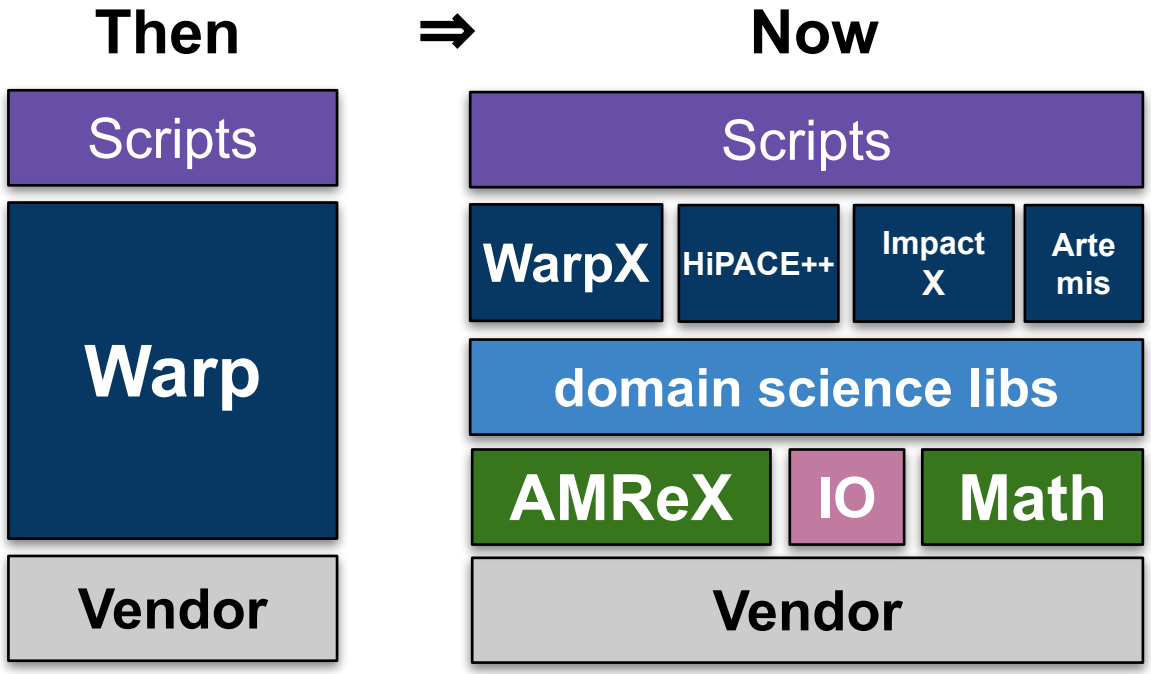


Desktop
to
HPC

We "graduated" from the "DIY" world

Deep, long-term collaborations:

- vendors, DOE ASCR, HPC centers, …
- own sub-libs/modules, contributions
- machines & deployed environments



Modular Software Architecture



Python: Modules, PICMI interface, Workflows



WarpX
fusion, accelerators

ImpactX
beam dynamics

…

HiPACE++
plasma-wakefield

ARTEMIS
microelectronics

mac OS

pyAMReX

PICSAR
QED Modules

ABLASTR: shared PIC



ML Frameworks
PyTorch, Tensorflow, …

AMReX

Containers, Communication, Solvers, Portability, Utilities

openPMD
diagnostics
I/O

Math

FFTs, lin. alg.

Desktop to HPC

CUDA, OpenMP, SYCL, HIP

MPI

WarpX: Open, international, neutral governance

Steering Committee (SC)

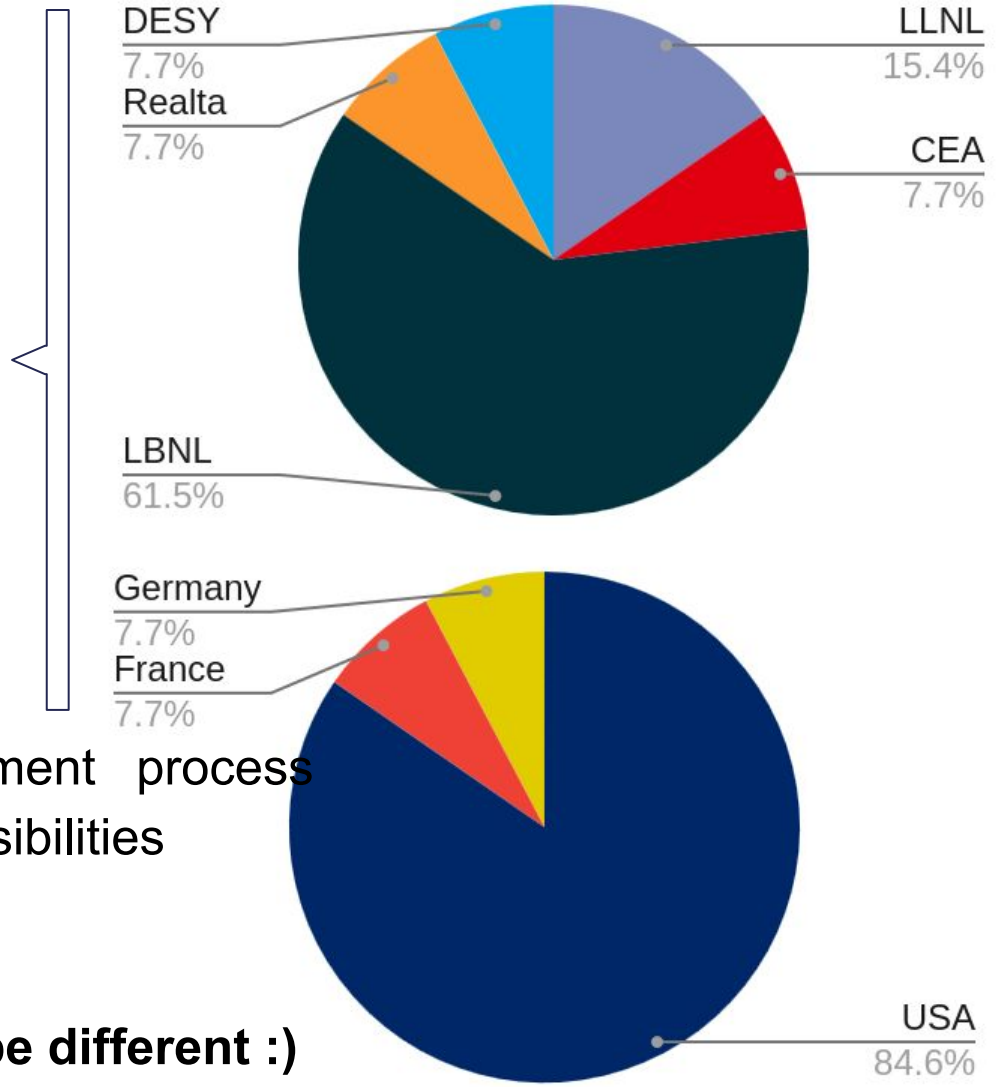
- admins (of last resort)

Technical Committee (TC)

- core governance body, where decisions are made
- maintainers, merge permissions

Contributors

- known contributors w/ triage & elevated testing privileges



Each role: current roster, role, decision process, appointment process
→ weekly developer meetings, expectations, privileges + responsibilities
Goal: sustainable, representative of work on the ground

This is documenting *our* governance structure – yours can be different :)

https://warpx.readthedocs.io/en/latest/governance.html

Participation in the HPSF Annual Developer Conference

Since 2025 an annual conference, dedicated to advancing community-driven solutions for performance, portability, and productivity in a rapidly evolving High Performance Computing (HPC) landscape.

HPSF project AMReX hosted a **Project Meeting** day, which HPSF project WarpX joined.

- connected across WarpX developer & AMReX project scopes
- discussed emerging technologies and needs



Additionally, contributed to **plenary talks** and **conference discussions**, e.g.,

- exchanged on HPC programming languages
  - C++: Kokkos, HPX, ISO C++, etc.
  - Python, standardization, workflows
- packaging & deployment
- project growth strategies
- project technical & community management
- hands-on hacked GPU CI with two distributed partners

In essence: Sharing, discovering, forming new collaborations.



HPSF TAC: A Wealth of Opportunity to Engage

Interaction w/ TAC projects

- **AMReX** – central partner for WarpX
  - upstream & downstream contribs
  - cross-project needs with other AMReX science applications
  - shared Python and AI/ML strategy
- **Spack** – central for packaging
  - WarpX ships Spack & Conda/Pixi
  - Spack is best-in-class for performance, GPU and HPC deployments, incl. **E4S**
  - Containers – growing adoption and **Apptainer** a very likely supported solution for us
- C++ Projects – harmonize & standardize
  - ISO C++, **Kokkos** & **HPX**
- In situ visualization – **Viskores** formerly VTK-m

WG services: GPU CI

- we collab'ed with U Oregon + Kitware
- our need: secure, GitHub-integrated, per-PR CI on real GPU hardware
- solution: maintainer-authorized (label/comment), GitLab mirrored, GH-status-indicated runners on recent Nvidia, AMD and Intel datacenter GPUs

Non-profit benefits

- The Linux Foundation is a non-profit
- Collective negotiations, e.g., for **Slack** licenses
- Shared resources & analysis, e.g., **project logo, health** and **planning** (LFX), outreach & **tutorials**
- Interaction w/ vendors (feedback, tests, bugs)



HPSF

HIGH PERFORMANCE
SOFTWARE FOUNDATION



ENVIRONMENT
MODULES

Environment Modules: from a 1991 tool to an HPSF project

Xavier Delaruelle, CEA

HPC Best Practices webinar, October 28, 2026

Change the shell environment on the fly

Load, unload, switch

A modulefile describes environment changes. module load applies them, module unload reverts them, in one atomic step.

One command for every shell

Same interface in bash, zsh, ksh, csh, tcsh, fish, cmd and PowerShell, and from Python, Perl, Ruby, R, CMake and Tcl scripts.

A script that prints shell code

modulecmd.tcl evaluates the modulefile and emits the matching shell commands, which the module function evals in the session.

```
$ cat /path/to/modulefiles/foo
#%Module
prepend-path PATH /path/to/apps/foo/bin

$ which foo
foo not found
$ module load foo
$ which foo
/path/to/apps/foo/bin/foo
$ module unload foo
$ which foo
foo not found
```

The front door to HPC software catalogs

Many applications, many versions

module avail lists the catalog, module load picks a version. Several versions of a tool coexist with no conflict.

One interface for every installer

Spack, EasyBuild and other installers all emit modulefiles. Users get the same command whatever produced the software stack.

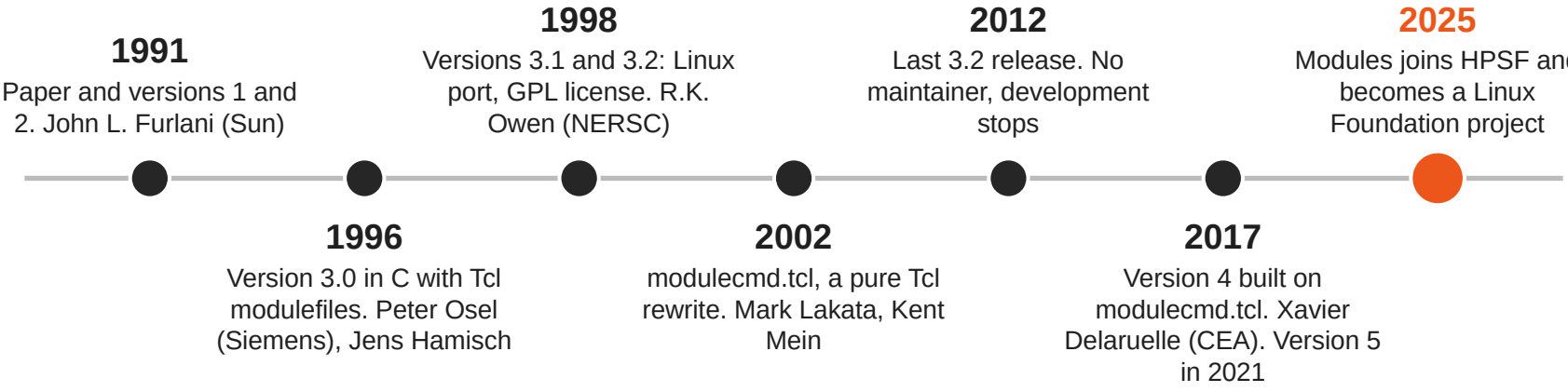
Dependencies resolved automatically

Requirements get loaded, conflicts unloaded, dependents reloaded. Module hierarchies and variants use the same mechanism.

More than software access

MPI profiles, data endpoints, development setups, collections: environment variables, but also aliases, shell functions and completion.

35 years old, revived by CEA in 2017



2 feature releases a year

~50 user-visible changes per release

24k+ non-regression tests, 99.5% coverage

v5.7.0 released September 2026

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Prominent features added since 2017

For users

Collections
v4.0 module save and restore a whole environment

Automated dependency handling
v4.2 requirements, conflicts, dependent reload

Advanced version specifiers
v4.4 module load foo@3.2:3.5

Module variants
v4.8 module load foo@2 +mpi toolchain=foss24a

Environment control

Initial environment and stash
v5.2 module reset, stash and stashpop

Extra specifiers
v5.3 module avail envvar:PATH require:foo

Module hierarchy
v5.6 spider and requiring via a module

Windows and PowerShell
v4.5 cmd support, PowerShell since v5.5

For sysadmins

Hide, forbid and tag modules
v4.6 control what users see and may load

source-sh
v4.6 turn a vendor shell script into a modulefile

Module cache
v5.3 and 63% faster loads in v5.7

Linting, logging and hook API
v5.2 to 5.7 quality checks, audit trail, site hooks

Why HPSF: one lab and one person is fragile

The situation in 2024

- CEA alone has carried the project since 2017, with a single maintainer.
- Name, logo and domain belonged to one organization: any hand-over would have been painful.
- A wide user base, but about two external contributors per release.

What a neutral foundation changes

- Linux Foundation Europe owns the project assets. No single employer controls the direction.
- Open governance written down: charter, roles, decisions taken in public.
- Visibility and new contributors through HPSF outreach, HPSFcon and sister projects such as Spack and Kokkos.

How it went

Dec 2024
Repository moved to the neutral envmodules GitHub organization; governance written

Jan 2025
Project proposal presented to the HPSF Technical Advisory Council

Sep 2025
Charter adopted; contributor agreement moves the assets to LF Europe

Nov 2025
HPSF announces Modules as an Established project

Feb 2026
First open TSC meeting; first project meeting at HPSFcon in March

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Under HPSF: charter, TSC and open meetings

Linux Foundation Europe

Legal home of the project. Holds its name, logo and domain. Every contribution carries a DCO sign-off.

High Performance Software Foundation

Directed fund of the Linux Foundation. Governing Board, Technical Advisory Council where Modules holds a seat, working groups, outreach team. Modules is at the Established stage.

Modules project

Technical Charter, adopted Sep 4, 2025

Mission, scope, roles and decision process. Maintainers, Collaborators and Former members listed in the GOVERNANCE file.

Technical Steering Committee

The TSC is made of the project maintainers. A majority vote adds or removes a maintainer. The TSC chair is the liaison with HPSF.

Decisions in the open

Quarterly TSC meetings on LFX Zoom, open to anyone, with agenda and notes on the GitHub wiki. The content of each release is agreed there.

One year in: more hands, more visibility

Governance that outlives its founder

Writing the charter made the project define roles and processes it had never written down.

A second maintainer

Adrien Cotte (AS+) joined the TSC in 2026. We look for new contributors who could become maintainers.

More contributions

A neutral home draws more interest: contributions have increased since the project joined.

A session at the HPSF annual conference

HPSFcon 2026 put us in touch with our user community. Their feedback now guides the roadmap.

Visibility and services

HPSF outreach: announcement, social media. LFX services: domain, mailing list, Zoom.



Get involved

github.com/envmodules/modules
envmodules.io, docs at modules.readthedocs.io
lists.hpsf.io/g/modules-interest, #modules:matrix.org
Quarterly TSC meetings open to all (project wiki)